Exercise Sheet

Cohomology on $\mathbb{R}^2 \setminus \{0\}$, S^2, T^2, and Gauss–Bonnet

A. Winding form on $\mathbb{R}^2 \setminus \{0\}$

A1. Let

$$\eta = \frac{-y\,dx + x\,dy}{x^2 + y^2} \qquad \text{on } \mathbb{R}^2 \setminus \{0\}.$$

Rewrite η in polar coordinates

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

and show that

$$\eta = d\theta.$$

A2. Compute directly that

$$d\eta = 0$$

on $\mathbb{R}^2 \setminus \{0\}$.

A3. Let $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$. Compute

$$\int_\gamma \eta.$$

A4. Explain why the result of A3 proves that η is not exact.

A5. Let $\gamma_n(t) = (\cos nt, \sin nt)$, $0 \le t \le 2\pi$, where $n \in \mathbb{Z}$. Compute

$$\int_{\gamma_n} \eta.$$

Interpret the answer in terms of winding number.

A6. Show that if γ is a closed curve in $\mathbb{R}^2 \setminus \{0\}$ which does not wind around the origin, then

$$\int_\gamma \eta = 0.$$

B. Area form on S^2

B1. Parametrize the unit sphere by

$$X(\theta, \phi) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi), \qquad 0 < \theta < 2\pi, \quad 0 < \phi < \pi.$$

Compute X_θ and X_ϕ.

B2. Show that the induced metric on S^2 is

$$ds^2 = d\phi^2 + \sin^2\phi\, d\theta^2.$$

B3. Deduce that the area form is

$$dA = \sin\phi\, d\theta \wedge d\phi.$$

B4. Show that dA is closed.

B5. Compute

$$\int_{S^2} dA.$$

B6. Explain why the result of B5 proves that dA is not exact.

B7. Conclude that

$$[dA] \neq 0 \in H^2_{\mathrm{dR}}(S^2).$$

B8. Explain informally why

$$H^1_{\mathrm{dR}}(S^2) = 0.$$

C. Local primitives on S^2

C1. Let

$$U_N = S^2 \setminus \{N\}, \qquad U_S = S^2 \setminus \{S\}.$$

Define

$$\alpha_N = (1 - \cos\phi)\, d\theta \qquad \text{on } U_N.$$

Compute $d\alpha_N$.

C2. Define

$$\alpha_S = -(1 + \cos\phi)\, d\theta \qquad \text{on } U_S.$$

Compute $d\alpha_S$.

C3. Show that $d\alpha_N$ and $d\alpha_S$ both agree with the area form up to orientation convention.

C4. On $U_N \cap U_S$, compute

$$\alpha_N - \alpha_S.$$

C5. Explain why the formula

$$\alpha_N - \alpha_S = 2\, d\theta$$

shows that the obstruction to global exactness of dA is related to winding.

C6. Compare this with the winding-form example on $\mathbb{R}^2 \setminus \{0\}$.

D. Cohomology of the torus T^2

D1. Let

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2.$$

Explain why the forms dx and dy on $\mathbb{R}^2$ descend to well-defined 1-forms on T^2.

D2. Show that

$$d(dx) = 0, \qquad d(dy) = 0.$$

D3. Let

$$\gamma_x(t) = (t, 0) \mod \mathbb{Z}^2, \qquad \gamma_y(t) = (0, t) \mod \mathbb{Z}^2, \qquad 0 \leq t \leq 1.$$

Compute

$$\int_{\gamma_x} dx, \qquad \int_{\gamma_x} dy, \qquad \int_{\gamma_y} dx, \qquad \int_{\gamma_y} dy.$$

D4. Explain why dx is not exact.

D5. Explain why dy is not exact.

D6. Show that $[dx]$ and $[dy]$ are linearly independent in

$$H^1_{\mathrm{dR}}(T^2).$$

D7. Conclude that

$$H^1_{\mathrm{dR}}(T^2) \cong \mathbb{R}^2.$$

D8. Explain why x and y are not globally single-valued functions on T^2, even though upstairs on $\mathbb{R}^2$ one has

$$dx = d(x), \qquad dy = d(y).$$

E. The top class on T^2

E1. Show that

$$d(dx \wedge dy) = 0.$$

E2. Compute

$$\int_{T^2} dx \wedge dy$$

using the square fundamental domain $[0,1] \times [0,1]$.

E3. Explain why $dx \wedge dy$ is not exact.

E4. Conclude that

$$[dx \wedge dy] \neq 0 \in H^2_{\mathrm{dR}}(T^2).$$

E5. Verify the wedge-product relations

$$[dx] \wedge [dx] = 0, \qquad [dy] \wedge [dy] = 0,$$

$$[dx] \wedge [dy] = [dx \wedge dy], \qquad [dy] \wedge [dx] = -[dx \wedge dy].$$

E6. Explain in words why the top-degree class on the torus comes from wedging the two basic winding classes.

F. Curvature form and Gauss–Bonnet

F1. Let M be an oriented Riemannian surface with Gaussian curvature K and area form dA. Explain why

$$K\,dA$$

is automatically closed.

F2. Suppose locally there is a connection 1-form ω_{12} such that

$$d\omega_{12} = K\,dA.$$

Explain why this means the curvature form is locally exact.

F3. State the Gauss–Bonnet theorem for a compact oriented surface.

F4. For the unit sphere, use $K \equiv 1$ to compute

$$\int_{S^2} K\,dA.$$

F5. Verify that your answer in F4 agrees with

$$2\pi\chi(S^2).$$

F6. For the flat torus with metric

$$ds^2 = dx^2 + dy^2,$$

show that

$$K \equiv 0.$$

F7. Deduce that

$$\int_{T^2} K\, dA = 0.$$

F8. Verify that your answer in F7 agrees with

$$2\pi\chi(T^2).$$

F9. Explain why

$$[dA] \neq 0 \text{ on } T^2 \qquad \text{but} \qquad [K\, dA] = 0$$

for the flat torus.

G. Euler class and the sphere transition function

G1. Define stereographic coordinates on $S^2 \setminus \{N\}$ and $S^2 \setminus \{S\}$ by

$$z = \frac{X + iY}{1 - Z}, \qquad w = \frac{X - iY}{1 + Z}.$$

G2. Show that on the overlap,

$$w = \frac{1}{z}.$$

G3. Compute the derivative

$$\frac{dw}{dz}.$$

G4. Restrict to the equator, where $z = e^{i\theta}$, and show that

$$\frac{dw}{dz} = -e^{-2i\theta}.$$

G5. Explain why this transition function winds twice around the circle as θ goes from 0 to 2π.

G6. Explain why this implies that the tangent bundle TS^2 has Euler number 2.

G7. Compare this with

$$\frac{1}{2\pi}\int_{S^2} K\, dA.$$

H. Synthesis questions

H1. What is the common idea behind the following three facts?

$$\eta = d\theta \text{ locally on } \mathbb{R}^2 \setminus \{0\},$$

$$dA = d\alpha_N = d\alpha_S \text{ locally on } S^2,$$

$$dx = d(x), \quad dy = d(y) \text{ upstairs on } \mathbb{R}^2.$$

H2. Explain in your own words the difference between:

$$\text{closed} \qquad \text{and} \qquad \text{exact}.$$

H3. Explain in your own words why S^2 has nontrivial H^2 but trivial H^1.

H4. Explain in your own words why T^2 has nontrivial H^1 and nontrivial H^2.

H5. Explain in your own words why Gauss–Bonnet is a statement about the curvature form $K\,dA$, not about the area form dA alone.

H6. Write one paragraph explaining the slogan

$$\boxed{\text{local primitives may exist, but a global primitive may fail to exist.}}$$

using examples from $\mathbb{R}^2 \setminus \{0\}$, S^2, and T^2.